

# Pravin G

📍 Chennai, Tamilnadu | ✉ Pravin.esd@gmail.com | ☎ +918248373910

## Embedded Linux Engineer | BSP & Device Driver Developer

**3+ years** in end-to-end embedded software development, specializing in **Linux BSP**, **device drivers**, **post-silicon validation**, and **Yocto-based system integration**.

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### Technical Skills

#### Embedded Linux & BSP

- ✓ **Linux Kernel:** 4.x, 5.x (Driver Development, Debugging, Porting)
- ✓ **BSP/Yocto:** Porting & Customization (Dunfell, Kirkstone) | U-Boot | Device Trees
- ✓ **Board Support:** NXP i.MX (6ULL, 8MQuad, 8MNano), Toradex, Raspberry Pi
- ✓ **Device Drivers:** SPI, I2C, UART, SDIO, MISC, Input, IIO

#### Post-Silicon & Validation

- ✓ Intel GPU Validation (ATSM, GNR-WS) | Hypervisor/ESXi | KVM | SRIOV
- ✓ Defect Root-Causing | Log Analysis | HSD/JIRA Tracking

#### Programming & Tools

- ✓ **Languages:** C, C++, Python, Bash
  - ✓ **Tools:** Git, JIRA, VMware, Visual Studio, MobaXterm
  - ✓ **Protocols:** SPI, I2C, UART, eSPI
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### Professional Experience

**Engineer | L&T Technology Services, Chennai (Feb 2023 – Present)**

#### Embedded & Platform Software BU

- **ESXi Device Driver Development (Intel Collaboration)**
  - Fixed **priority defects** in ESXi networking drivers by analyzing kernel logs.
  - Reproduced issues in **VMware Horizon VDI** and resolved **NIC card compatibility bugs**.
- **Post-Silicon Validation (Intel ATSM GPU)**
  - Validated **3D workloads (Unigine, Furmark)** for 48+ hours to test GPU Stress & stability.

- Debugged **ESXi virtualization issues** in VDI environments using **JIRA & kernel traces**.
- Ensured BKC compliance across **Ubuntu, SLES, RHEL** for driver releases.
- **GNR-WS Workstation Validation**
  - Developed **Python scripts** to debug **eSPI (Enhanced Serial Peripheral Interface)** issues.
  - Analyzed **SAF/VW channel flows** for peripheral communication.

**Embedded Software Engineer | NEE VEE Technologies (Feb 2022 – Feb 2023)**

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### **Key Projects**

#### **1. Yocto BSP Porting – NXP i.MX8MQuad & i.MX8MNano**

- Ported Linux 5.15 on custom boards using Yocto (Dunfell/Kirkstone).
- Developed SPI/I2C/SDIO drivers for Wi-Fi/BT modules.
- Debugged boot-time issues in U-Boot and kernel initialization.

#### **2. Raspberry Pi 4 – Kernel Module Development**

- Ported sensor drivers (Si7006, MMA8652) using MISC framework.
- Designed sysfs interfaces for user-space control.

#### **3. Qt5 HMI Application Deployment (Yocto)**

- Created custom meta-layer for Qt5 on Raspberry Pi 4.
- Reduced boot time by 20% via filesystem optimization.

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### **Education**

**Bachelor's in Computer Science Engineering** | Dr.MGR Educational and Research Institute of Technology, Chennai | **2019-2022**